ECLT5930B/SEEM5740B Engineering Economics and Cost Management

Chapter 10
Evaluate projects with the benefit-cost ratio method

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Nature of Public Projects

- Public projects differ from private ones in many aspects.
  - Purpose: provide service at low cost, protect lives and properties, etc.
  - Project Life: relatively long
  - Benefits: often non–monetary and difficult to quantify
  - Conflict of Interests: common among different agencies of the government
  - Effect of Politics: can be a serious and often critical factor

Nature of Public Projects

- Decisions for a public project must balance the interests of various groups:
  - the public, who pays for the project through taxes
  - the decision makers, who have different political interests
  - the operators, who manage the project

Nature of Public Projects

	Private	Public
Purpose	Provide goods or services at a profit; maximize profit or minimize cost	Protect health; protect lives and property; provide services (at no profit); provide jobs
Sources of capital	Private investors and lenders	Taxation; private lenders
Method of financing	Individual ownership; partnerships; corporations	Direct payment of taxes; loans without interest; loans at low interest; self-liquidating bonds; indirect subsidies; guarantee of private loans
Multiple purposes	Moderate	Common (e.g., reservoir project for flood control, electrical power generation, irrigation, recreation, education)
Project life	Usually relatively short (5 to 10 years)	Usually relatively long (20 to 60 years)
Relationship of suppliers of capital to project	Direct	Indirect, or none
Nature of benefits	Monetary or relatively easy to equate to monetary terms	Often nonmonetary, difficult to quantify, difficult to equate to monetary terms
Beneficiaries of project	Primarily, entity undertaking project	General public
Conflict of purposes	Moderate	Quite common (dam for flood control versus environmental preservation)
Conflict of interests	Moderate	Very common (between agencies)
Effect of politics	Little to moderate	Frequent factors; short-term tenure for decision makers; pressure groups; financial and residential restrictions; etc.
Measurement of efficiency	Rate of return on capital	Very difficult; no direct comparison with private projects

The Benefit–Cost (BC) Ratio Method

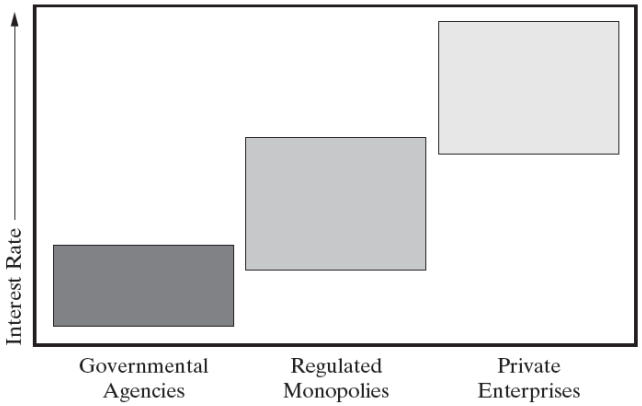
- A popular method for evaluating public project.
- Essentially, it involves calculating the ratio of (discounted) project benefits to (discounted) project costs.
- Questions
  - Whose benefits are we considering?
  - Who bears the costs?
- It is also necessary to consider disbenefits, or negative consequences, of a project to the public.

Example: Building a Third Runway at the HKIA

- Questions
  - What are the benefits?
  - What are the costs?
  - What are the disbenefits?

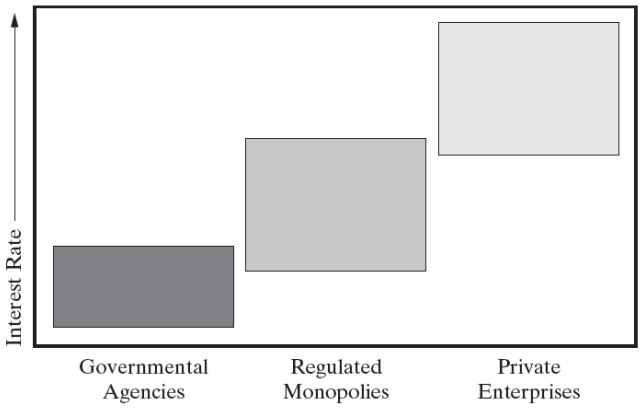
Evaluating Public Projects: Interest Rate

- However, the rationale for the use of interest rates is somewhat different.
  - In private projects, the choice of an interest rate is intended to lead to a selection of projects that maximize profit.
  - In public projects, the choice of an interest rate is intended to determine how available funds should be allocated among competing projects to achieve social goals.



Evaluating Public Projects: Interest Rate

- Factors affecting the choice of an interest rate in public projects include
  - interest rate on borrowed capital,
  - opportunity cost of capital to the governmental agency, and
  - opportunity cost of capital to taxpayers.



The BC Ratio Method: PW Formulation

- Notations
  - *B* = benefits of proposed project
  - *I* = initial investment in the proposed project
  - MV = market value at the end of useful life
  - O&M = operating and maintenance costs of the proposed project
- Conventional BC ratio with PW:

$$BC_c = \frac{PW(benefit\ of\ proposed\ project)}{PW(cost\ of\ proposed\ project)} = \frac{PW(B)}{I - PW(MV) + PW(O\&M)}$$

- Modified BC ratio with PW:

$$BC_M = \frac{PW(B) - PW(O\&M)}{I - PW(MV)}$$

The BC Ratio Method: AW Formulation

- Conventional BC ratio with AW:

$$BC_c = \frac{AW(benefit\ of\ proposed\ project)}{AW(cost\ of\ proposed\ project)} = \frac{AW(B)}{CR + AW(O\&M)}$$

  – CR is the capital recovery amount, i.e., the equivalent annual cost of the initial investment and salvage value

- Modified BC ratio with AW:

$$BC_M = \frac{AW(B) - AW(O\&M)}{CR}$$

The BC Ratio Method: Decision Rule

- If the BC ratio exceeds 1, then the project is acceptable.
- Question: Can the different BC ratio methods give different conclusions?
  - The conventional BC ratio with PW should give the same conclusion as the one with AW, since PW and AW are equivalent measures of worth.
  - The same applies to modified BC ratio with PW and modified BC ratio with AW.
  - It turns out that the conventional and modified BC ratio methods will give the same conclusion as well.

Equivalence of Conventional and Modified BC Methods

- Suppose that

$$BC_c = \frac{PW(B)}{I - PW(MV) + PW(O\&M)} > 1$$

$$I - PW(MV) > 0$$

- Then

$$PW(B) > I - PW(MV) + PW(O\&M)$$

$$PW(B) - PW(O\&M) > I - PW(MV)$$

$$BC_M = \frac{PW(B) - PW(O\&M)}{I - PW(MV)} > 1$$

Example 10.1: Runway Extension

- Suppose that we are interested in extending the runways of a city airport
- Interest rate: 10%; study period: 20 years.
- The costs and benefits are listed as follows

Cost	
Farmland reclamation	$350,000
Construction	$600,000
Terminal building	$250,000
Annual maintenance	$97,500
Air traffic controllers	$100,000

Benefit	
Rental from airlines	$325,000
Airport taxes	$65,000
Convenience	$50,000
Tourism	$50,000

- Question: Is the project acceptable?

Example 10.1: Runway Extension

- We compute the conventional BC ratio with AW:

$$BC_c = \frac{AW(B)}{CR + AW(O\&M)} = \frac{490{,}000}{1{,}200{,}000(A/P, 10\%, 20) + 197{,}500} = 1.448 > 1$$

- Alternatively, we can compute the modified BC ratio with AW

$$BC_M = \frac{AW(B) - AW(O\&M)}{CR} = \frac{490{,}000 - 197{,}500}{1{,}200{,}000(A/P, 10\%, 20)} = 2.075 > 1$$

Disbenefits in the BC Ratio Method

- D be the amount of disbenefits.
- Conventional BC ratio with AW, benefits reduced by disbenefits:

$$BC_c = \frac{AW(benefits) - AW(disbenefits)}{AW(total\ costs)} = \frac{AW(B) - AW(D)}{CR + AW(O\&M)}$$

- Conventional BC ratio with AW, costs increased by amount of disbenefits:

$$BC_c = \frac{AW(benefits)}{AW(disbenefits) + AW(total\ costs)} = \frac{AW(B)}{CR + AW(D) + AW(O\&M)}$$

Example 10.1: Runway Extension

- Continuing the runway extension example, suppose that the noise pollution results in a disbenefit of $100,000 per year.
- Conventional BC ratio with AW, benefits reduced by disbenefits

$$BC_c = \frac{AW(B) - AW(D)}{CR + AW(O\&M)} = \frac{490{,}000 - 100{,}000}{1{,}200{,}000(A/P, 10\%, 20) + 197{,}500} = 1.152 > 1$$

- Conventional BC ratio with AW, costs increased by amount of disbenefits:

$$BC_c = \frac{AW(B)}{CR + AW(D) + AW(O\&M)}$$

$$= \frac{490{,}000}{1{,}200{,}000(A/P, 10\%, 20) + 100{,}000 + 197{,}500} = 1.118 > 1$$

News Reference: 3rd Runway of HK airport

In March 2012, the Executive Council ("ExCo"), in-principle and for planning purposes, approved the AAHK to adopt the 3RS as the future development direction for HKIA.[12] Subsequently, the AAHK commenced the Environmental Impact Assessment ("EIA") studies which analysed the potential environmental impact of the project across 12 aspects such as noise and fisheries.[13]

An exhibition and public forums were held in 2013,[9] and the EIA report was submitted to the Environmental Protection Department for consideration in April 2014.[9] The report was then available for a one-month public inspection,[14] and was subsequently approved with the Environmental Permit granted by the Director of Environmental Protection on 7 November 2014.[13] This was highly criticised by environmental groups such as Friends of the Earth for neglecting public and expert opinions.[15]

The ExCo further endorsed the expansion of the HKIA into a 3RS on 17 March 2015.[9] The project is expected to cost HK$141.5 billion, with funding coming from bank loans, bonds issuance, HKIA's operational surplus and end-users. An Airport Construction Fee of HK$180 per departing passenger was proposed,[16] but the AAHK subsequently lowered it, with 70% of departing passengers paying HK$90 or less as announced in September 2015.[4] The funding principle of "Joint Contribution and User-pay" remained unchanged.[4]

The endorsement of the ExCo sparked public outrage, as it left a lot of questions such as the use of airspace unanswered,[17] and this funding structure was said to have effectively bypassed the Legislative Council ("LegCo")[18] because Government guarantee, allocation of funds and passing of resolution by the LegCo were not required.[18] Members of Panel on Economic Development raised similar concerns, in which the Administration responded by saying that the AAHK was granted power under the Airport Authority Ordinance to make necessary financial arrangements for airport expansion and operation.[19] A "Subcommittee to Follow Up Issues Relating to the 3RS at the HKIA" was endorsed on 15 May 2015 upon the request of the Panel on Economic Development of LegCo.[19]

The draft Chek Lap Kok Outline Zoning Plan was approved and the reclamation was authorised by the Chief Executive-in-Council on 26 April 2016.[9] This was once again disapproved by Green Sense for ignoring the ecological, noise and airspace problems.[20] The construction of the 3RS and the collection of the Airport Construction Fee started on 1 August 2016,[21] and is expected to complete by 2023.

Interaction between government and different stakeholders [edit]

Between AAHK and government and the public [edit]

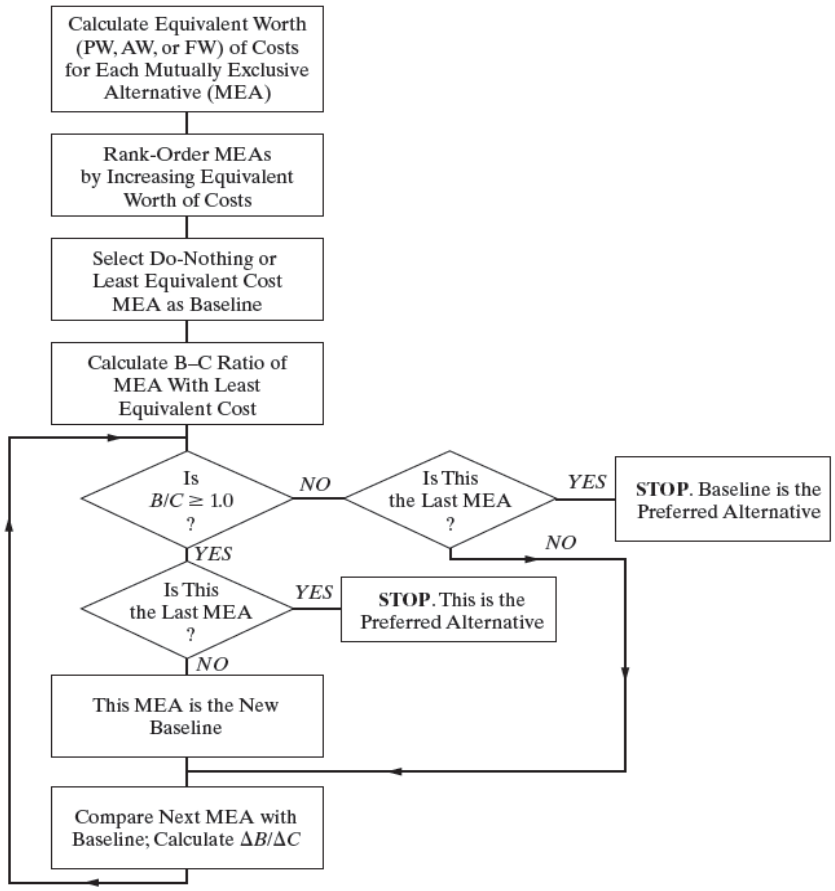
AAHK has been taking a leading role in introducing the 3RS Project, with great emphasis on the urgency of the current problem regarding insufficiency of the existing 2RS in meeting rising future air traffic demand. It is claimed that, by 2016 or 2017, the existing 2RS will reach its maximum practical capacity of 68 ATMs per hour, or 420,000 ATMs per year.[4] In face of the fierce competition from neighbouring airports who have been actively planning for their expansion projects, AAHK expected that the 3RS will be able to handle over 100 million passengers, nearly 9 million tonnes of cargo and 607,000 ATMs which would enable Hong Kong to maintain the competitive advantage as an aviation hub in attracting foreign businesses.[4] It is estimated that by 2030 the 3RS would have generated HK $167 million,[7] and will further bring economic benefits of $455 billion over 50 years.

The proposal has attracted widespread public opposition, and has aroused great concern on whether the infrastructure project which involved the highest cost since 1997 is justified by its alleged urgency and effectiveness. Nevertheless, despite public challenge over the enormous economic benefits as claimed, the government has attached great importance to the 3RS project and has largely agreed that there are strong justifications and urgency to proceed with the implementation.[22] The Secretary for Transport and Housing Bureau Anthony Cheung has also publicly expressed the government view that such a project would be endorsed and recognized its potential economic contribution to Hong Kong's GDP.[18]

Comparison of mutually exclusive projects using the BC ratio method

- Incremental analysis procedure based on the BC ratio method
  - Rank the alternatives in increasing order of total equivalent worth of costs. The do–nothing alternative is the initial baseline alternative.
  - Compute the difference in benefits (ΔB) and costs (ΔC) between the next least equivalent worth alternative and the baseline alternative, and calculate the incremental BC ratio $\Delta B/\Delta C$.
  - If the ratio is at least 1, then the higher equivalent worth alternative becomes the new baseline. Otherwise, the last baseline alternative is maintained.
  - Repeat until all the alternatives have been considered.

Comparison of mutually exclusive projects using the BC ratio method



Example 10.2

- Consider the following three projects

	Project X	Project Y	Project Z
Capital investment	$8,500	$10,000	$12,000
Annual O&M cost	$750	$725	$700
Annual benefit	$2,150	$2,265	$2,500
Market value	$1,250	$1,725	$2,000

- Suppose that MARR is 10% per year, and the study period is 50 years.
- Question:
  - What are the conventional BC ratios of the projects? What are the modified BC ratios of the projects?

Example 10.2

- The present worth of the cost of the projects are

$$PW_{c,X}(10\%) = 8{,}500 + 750 \times (P/A, 10\%, 50) - 1{,}250 \times (P/F, 10\%, 50)$$
$$= 15{,}925.46$$

$$PW_{c,Y}(10\%) = 10{,}000 + 725 \times (P/A, 10\%, 50) - 1{,}750 \times (P/F, 10\%, 50)$$
$$= 17{,}173.55$$

$$PW_{c,Z}(10\%) = 12{,}000 + 700 \times (P/A, 10\%, 50) - 2{,}000 \times (P/F, 10\%, 50)$$
$$= 18{,}923.33$$

- The present worth of the benefit of the projects are

$$PW_{b,X}(10\%) = 2{,}150 \times (P/A, 10\%, 50) = 21{,}316.85$$
$$PW_{b,Y}(10\%) = 2{,}265 \times (P/A, 10\%, 50) = 22{,}457.05$$
$$PW_{b,Z}(10\%) = 2{,}500 \times (P/A, 10\%, 50) = 24{,}787.04$$

Example 10.2

- Project X requires the least capital investment,

$$BC_X = \frac{21{,}316.85}{15{,}925.46} = 1.3385$$

  which is acceptable

- The incremental project Δ(Y-X) yields a BC ratio of

$$BC_{\Delta(Y-X)} = \frac{22{,}457.05 - 21{,}316.85}{17{,}173.55 - 15{,}925.46} = 0.9137$$

  which is unacceptable

- The incremental project Δ(Z-X) yields a BC ratio of

$$BC_{\Delta(Z-X)} = \frac{24{,}787.04 - 21{,}316.85}{18{,}923.33 - 15{,}925.46} = 1.1576$$

  Therefore, project Z is selected as the final result

Drawbacks of the BC ratio method

- The required data might be hard to quantify.
- It ignores the problem of economic inequalities; i.e., one part of the population benefits at the expense of the other.
- It does not take any qualitative information into account.

- Chapter 10 of the course textbook.